MATH 415 - FALL 2020 - EXTRA PRACTICE PROBLEMS - WEEK 1

1. *Consider the following two linear equations:*

(a) $$2x_1 + 5x_2 = 2$$

(b) $$2x_1 + 5x_2 = 0.$$

If (x_1, x_2) and (x_1', x_2') are solutions to (a), is $(x_1 + x_1', x_2 + x_2')$ also a solution to (a)? If (x_1, x_2) and (x_1', x_2') are solutions to (b), is $(x_1 + x_1', x_2 + x_2')$ also a solution to (b)?

No. $(1, 0)$ and $(0, \frac{2}{5})$ are solutions to (a), but $(1, \frac{2}{5})$ is not.
Yes. Let (x_1, x_2) and (x_1', x_2') be solutions to (b). Then

$$2(x_1 + x_1') + 5(x_2 + x_2') = 2x_1 + 5x_2 + 2x_1' + 5x_2' = 0 + 0 = 0.$$

2. *Consider the following system:*

(c)
$$\begin{aligned} a_{1,1}x_1 + \cdots + a_{1,n}x_n &= b_1 \\ \vdots \qquad\qquad \vdots \qquad &= \vdots \\ a_{m,1}x_1 + \cdots + a_{m,n}x_n &= b_m \end{aligned}$$

Answer the following questions!

(1) *If $(x_1, \ldots, x_n)$ and $(x_1', \ldots, x_n')$ are solutions to (c), is $(x_1 + x_1', \ldots, x_n + x_n')$ also a solution to (c)?*
(2) *What conditions on $b_1, \ldots, b_m$ can you impose such that (1) becomes true?*

(1) No. Because in (1) we already saw a system for which this statement fails.
(2) Suppose we have $b_1 = \cdots = b_m = 0$. Let $(x_1, \ldots, x_n)$ and $(x_1', \ldots, x_n')$ be solutions to (c). Then for each $j = 1, \ldots, m$ we have

$$\begin{aligned} a_{j,1}(x_1 + x_1') + \cdots + a_{j,n}(x_n + x_n') & \\ = a_{j,1}x_1 + \cdots + a_{j,n}x_n + a_{j,1}x_1' + \cdots + a_{j,n}x_n' & \\ = 0 + 0 = 0 = b_j. & \end{aligned}$$

Therefore under this assumption, $(x_1 + x_1', \ldots, x_n + x_n')$ is also a solution to (c).

3. *What is the largest possible number of pivots a 4×6 matrix can have?*

Each row can have at most one leading entry and hence at most one pivot.

4. *What is the largest possible number of pivots a 6×4 matrix can have?*

Again 4. Each column has at most one pivot.

5. *How many solutions does a consistent linear system of 3 equations and 4 unknowns have?*

Solutions will be posted at the end of Week 2.

The augmented matrix of this system has 3 rows and 4 columns. Hence it has at most 3 pivots. So one variable can not correspond to a pivot column. Hence the system has one free variable. Any consistent system with a free variable has infinitely many solutions.

6. *Just using elementary row operations, can you transform* $\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$ *into* $\begin{bmatrix}0 & 0\\ 0 & 0\end{bmatrix}$*?*

No. Both matrices are in reduced echelon form. Since the reduced echelon form of a matrix is unique and you can reduce every matrix into its reduced echelon form by just using elementary row operations, you can not transform $\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$ into $\begin{bmatrix}0 & 0\\ 0 & 0\end{bmatrix}$.

7. *Let* $A = [a_{ij}]_{3\times 4}$*, and let* $B = [b_{ij}]_{3\times 4}$ *be an echelon form of* A*.*

7.1. *Is it true that, if* $a_{11} = 0$*, then* $b_{11} = 0$*?*

False, consider $A = \begin{bmatrix}0 & 0 & 1 & 0\\ 0 & 1 & 0 & 0\\ 1 & 0 & 0 & 0\end{bmatrix}$.

7.2. *Is it true that, if* A *has a column of zeros, then* B *also has a column of zeros?*

True, since row operations do not change a column of zeros.

7.3. *Suppose* B *has a row of zeros. What can you say about rows of* A*?*

Each row of B is a linear combination of rows of A with at least one non-zero coefficient. Therefore, a linear combination of rows of A with at least one non-zero coefficient is equal to zero.

8. *Describe all* 2×2*-matrices that are of reduced echelon form!*

A 2×2-matrices is of reduced echelon form if and only if it is exactly one of the following matrices:

- $\begin{bmatrix}0 & 0\\ 0 & 0\end{bmatrix}$,
- $\begin{bmatrix}1 & a\\ 0 & 0\end{bmatrix}$, where a is some number,
- $\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$,
- $\begin{bmatrix}0 & 1\\ 0 & 0\end{bmatrix}$.

9. *True or false? If true, explain why! If false, give a counterexample.*

(1) *There is no more than one pivot position in any row.*

True. There can only be one leading entry in each row. So there can only be one pivot in each row.

(2) *There is no more than one pivot position in any column.*

True. In the reduced echelon form there is only a single non-zero entry in a column with a pivot. Therefore there is at most one pivot in each column.

(3) *There cannot be more free variables than pivot variables.*

False. Look at the equation

$$x_1 + x_2 + x_3 = 0.$$

The variables x_2 and x_3 are free, while x_1 is a pivot variable.

(4) *There is a linear system that has exactly two solutions.*

False. Every linear system has either no solution, exactly one solution or infinitely solutions. But never exactly *two (or exactly three) solutions.*

(5) *If the augmented matrix of a linear system has two identical rows, the linear system is necessarily inconsistent.*

This is false. The linear system

$$x_1 + x_2 = 0$$
$$x_1 + x_2 = 0$$

has infinitely many solutions.